EXPERIMENT - 2

AIM: Convert the ER diagram into a relational schema. Write complete mySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

SOURCE CODE:

Table	Primary key	Foreign keys
CUSTOMER	CUSTOMERID	-
CUSTOMERPHONE	(CustomerID, Phone)	CUSTOMERID → CUSTOMER
ADDRESS	AddressID	CustomerID → CUSTOMER
SELLER	SellerID	-
CATEGORY	CategoryID	-
PRODUCT	ProductID	CategoryID → CATEGORY
PRODUCT_SELLER	(ProductID, SellerID)	ProductID → PRODUCT, SellerID → SELLER
ELECTRONICS	ProductID	ProductID → Product
CLOTHING	ProductID	ProductID → Product
GROCERY	ProductID	ProductID → Product
ORDERS	OrderID	CustomerID → CUSTOMER
ORDERITEM	(OrderID, ProductID)	OrderID → ORDERS, ProductID → PRODUCT
PAYMENT	PaymentID	OrderID → ORDERS
DELIVERY	DeliveryID	OrderID → ORDERS

SQL Queries:

→ Schema Creation

```
CREATE DATABASE ECOMMERCE_ER;
USE ECOMMERCE_ER;

CREATE TABLE CUSTOMER(
        CustomerID      INT PRIMARY KEY,
        First_Name      VARCHAR(30) NOT NULL,
        Last_Name       VARCHAR(30) NOT NULL,
        Email           VARCHAR(100) NOT NULL UNIQUE,
        Date_of_Birth   DATE NOT NULL
);

CREATE TABLE CUSTOMER_PHONE(
        CustomerID  INT,
        Phone       VARCHAR(15),
        PRIMARY KEY (CustomerID, Phone),
        FOREIGN KEY (Customer_ID) REFERENCES CUSTOMER
        (CustomerID) ON DELETE CASCADE
);

CREATE TABLE ADDRESS(
        AddressID       INT PRIMARY KEY,
        House_No        VARCHAR(10),
        Street          VARCHAR(50),
        City            VARCHAR(30) NOT NULL,
        State           VARCHAR(30) NOT NULL,
        PIN_Code        VARCHAR(30) NOT NULL,
        Address_Type    VARCHAR(20),
```

```
    CustomerID   INT NOT NULL,
    FOREIGN KEY(CustomerID) REFERENCES CUSTOMER
        (CustomerID)  ON DELETE CASCADE
);

CREATE TABLE SELLER (
    SellerID   INT PRIMARY KEY,
    Seller_Name  VARCHAR(50) NOT NULL, ~~UNIQUE,~~
    Email      VARCHAR(100) NOT NULL, UNIQUE,
    Phone      VARCHAR(15),
    GSTIN      VARCHAR(20) NOT NULL UNIQUE
);

CREATE TABLE CATEGORY (
    CategaryID   INT PRIMARY KEY,
    Category_Name  VARCHAR(50) NOT NULL UNIQUE,
    Description   VARCHAR(150)
);

CREATE TABLE PRODUCT (
    ProductID      INT PRIMARY KEY,
    Product_Name   VARCHAR(100) NOT NULL,
    Price          DECIMAL(10,2) NOT NULL,
    Description    VARCHAR(200),
    Stock_Quantity  INT NOT NULL DEFAULT 0,
    Brand          VARCHAR(50),
    CategoryID     INT,
    FOREIGN KEY(Category_ID) REFERENCES CATEGORY
```

```
        (CategoryID) ON DELETE SET NULL
);

CREATE TABLE PRODUCT_SELLER(
    ProductID INT,
    SellerID INT,
    PRIMARY KEY (ProductID, SellerID),
    FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
        ON DELETE CASCADE,
    FOREIGN KEY(SellerID) REFERENCES SELLER(SellerID)
        ON DELETE CASCADE
);

CREATE TABLE ELECTRONICS(
    ProductID INT PRIMARY KEY,
    Warranty_Period VARCHAR(30),
    Model_Number VARCHAR(50),
    FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
        ON DELETE CASCADE
);

CREATE TABLE CLOTHING(
    ProductID INT PRIMARY KEY,
    Size VARCHAR(10),
    Colour VARCHAR(30),
    Material VARCHAR(30),
    FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE GROCERY (
        ProductID    INT PRIMARY KEY,
        Expiry_Date  DATE,
        Weight       DECIMAL(8,2),
        FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
              ON DELETE CASCADE
);

CREATE TABLE ORDERS (
        OrderID      INT PRIMARY KEY,
        Order_Date   DATE NOT NULL,
        Order_Status VARCHAR(20) NOT NULL,
        Total_Amount DECIMAL(10,2) NOT NULL,
        CustomerID   INT NOT NULL,
        FOREIGN KEY(CustomerID) REFERENCES CUSTOMER
              (CustomerID) ON DELETE CASCADE

);

CREATE TABLE ORDERITEM(
        OrderID    INT,
        ProductID  INT,
        Quantity   INT NOT NULL,
        Unit_Price DECIMAL(10,2) NOT NULL,
        Subtotal   DECIMAL(10,2) NOT NULL

        PRIMARY KEY(OrderID, ProductID),
        FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
              ON DELETE CASCADE,
        FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
              ON DELETE CASCADE
);
```

```
CREATE TABLE PAYMENT(
        PaymentID           INT PRIMARY KEY,
        Payment-Date        DATE NOT NULL,
        Amount              DECIMAL(10,2) NOT NULL,
        Payment-Method      VARCHAR(30) NOT NULL,
        Payment-Status      VARCHAR(20) NOT NULL,
        TransactionID       VARCHAR(30) UNIQUE,
        OrderID             INT NOT NULL UNIQUE,
        FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
                ON DELETE CASCADE
);

CREATE TABLE DELIVERY(
        DeliveryID          INT PRIMARY KEY,
        Delivery-Date       DATE
        Delivery-Status     VARCHAR(20) NOT NULL,
        Tracking-ID         VARCHAR(30) UNIQUE,
        Delivery-Partner    VARCHAR(50),
        OrderID             INT NOT NULL UNIQUE,
        FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
                ON DELETE CASCADE
);
```

→ Sample Data

```
USE ECOMMERCE_ER;

INSERT INTO CUSTOMER VALUES
(1, 'Sweksha', 'Singh', 'swek6105@gmail.com', '2006-04-15');

INSERT INTO CUSTOMER_PHONE VALUES
(1, '9939970621'),
(1, '9873921570');

INSERT INTO ADDRESS VALUES
(401, '307', 'Ganahinagar', 'Rochas', 'Bihar', '821112',
    'Home', 1);

INSERT INTO SELLER VALUES
(101, 'TechStore India', 'tech@gmail.com', '9123456789',
    'GST1234567');

INSERT INTO CATEGORY VALUES
(201, 'Electronics', 'Electronic gadgets and accessories');

INSERT INTO PRODUCT VALUES
(301, 'Gaming Laptop', 75000.00, '15.6 inch, RTX graphics',
    10, 'ASUS', 201);

INSERT INTO ELECTRONICS VALUES
(301, '2 Years', 'ASUS-TUF-F15');

INSERT INTO PRODUCT_SELLER VALUES
(301, 101);

INSERT INTO ORDERS VALUES
(501, '2026-08-14', 'Placed', 75000.60, 1);
```

```
INSERT INTO ORDERITEM VALUES
(501, 301, 1, 75000.00, 75000.00);

INSERT INTO PAYMENT VALUES
(601, '2026-08-14', 75000.00, 'UPI', 'Paid', 'TXN123', 501);

INSERT INTO DELIVERY VALUES
(701, NULL, 'Processing', 'TRK123', 'Delivery', 501);
```

⇒ Referential Integrity Violation

```
INSERT INTO ORDERS VALUES
(502, '2026-08-14', 'Placed', 50000.00, 999);
```

⇒ ON DELETE SET NULL Demonstration

```
DELETE FROM CATEGORY WHERE CategoryID = 201;
```

⇒ ON DELETE CASCADE Demonstration

```
DELETE FROM CUSTOMER WHERE CustomerID = 1;
```

OUTPUT:

⇒ Output 1 - SHOW TABLES

```
SHOW TABLES;
```

Tables_in_ECOMMERCE_ER
ADDRESS
CATEGORY
CLOTHING
CUSTOMER
CUSTOMER_PHONE
DELIVERY
ELECTRONICS
GROCERY
ORDERITEM
ORDERS
PAYMENT
PRODUCT
PRODUCT_SELLER
SELLER

⇒ Output 2 - Sample Data

```
SELECT * FROM CUSTOMER;
```

CustomerID	First_Name	Last_Name	Email	DOB
1	Sweksha	Singh	swek6105@gmail.com	2006-09-15

```
SELECT * FROM PRODUCT;
```

ProductID	Product_Name	Price	Stock_Quantity	Brand	CategoryID
301	Gaming Laptop	75000.00	10	ASUS	201

```
SELECT * FROM ORDERS;
```

OrderID	Order_Date	Order_Status	Total_Amount	CustomerID
501	2026-08-14	Placed	75000.00	1

⇒ Output 3 - Referential Integrity Violation

```
INSERT INTO ORDERS VALUES (502, '2026-08-14', 'Placed',
    50000.00, 999);
-- Since CustomerID = 999 does not exist in CUSTOMER:
ERROR 1452 (23000): Cannot add or update a child row:
a foregin key constraint fails ('ECOMMERCE_ER'.'ORDERS',
CONSTRAINT 'ORDERS_ibfk_1' FOREIGN KEY ('CustomerID')
REFERENCES 'CUSTOMER' ('CustomerID') ON DELETE CASCADE)
```

Reason: CustomerID 999 does not exist in CUSTOMER.
~~Satisfi~~.

⇒ Output 4 - ON DELETE SET NULL (CATEGORY → PRODUCT)

Before deletion:

ProductID	Product_Name	CategoryID
301	Gaming Laptop	201

```
DELETE FROM CATEGORY WHERE CategoryID = 201;
```

After deletion:

ProductID	Product_Name	CategoryID
301	Gaming Laptop	NULL

~~ON~~ +

⇒ Output 5 - ON DELETE CASCADE (CUSTOMER → everything downstream)

```
-- Before deleting CustomerID = 1, related rows exist
   in ORDERS, ORDERITEM, PAYMENT, DELIVERY, ADDRESS
   and CUSTOMER_PHONE (1 row each / 1 order item).
DELETE FROM CUSTOMER WHERE CustomerID = 1;
Query OK, 1 row affected.
```

Table	Row count after DELETE
ORDERS	0
ORDERITEM	0
PAYMENT	0
DELIVERY	0
ADDRESS	0
CUSTOMER_PHONE	0

